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# Determinants of Solar Panel Adoption Intention Among Island Communities: Case Study in Pulau Tidung, Indonesia

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**Abstract:** This study investigates the determinants of household intention to adopt solar photovoltaic (PV) systems in an island community using a stimulus organism response (SOR)-based framework informed by selected concepts from the Theory of Planned Behavior (TPB) and Technology Acceptance Model (TAM). Data were collected from 100 households in Pulau Tidung, Seribu Islands, Indonesia, using purposive sampling and analyzed through partial least squares structural equation modeling (PLS-SEM) to examine the effects of economic, environmental, social, regulatory, technical, and market stimuli on attitude and adoption intention. The results indicate that attitude toward solar PV exhibited the largest positive association with adoption intention ($\beta = 0.827$, $p < 0.001$), while economic stimulus was positively associated with attitude ($\beta = 0.411$, $p = 0.018$). Regulatory stimulus directly increases adoption intention ($\beta = 0.349$, $p = 0.042$), whereas technical barriers significantly reduce it ($\beta = -0.306$, $p = 0.036$). Environmental, social, and market stimuli do not exhibit statistically significant effects. The model explains 95.5% of the variance in adoption intention ($R^2 = 0.955$), indicating substantial explanatory power within the study context. These findings demonstrate that favorable attitudes, supportive regulatory conditions, and reduced technical barriers are central to strengthening household solar PV adoption intentions in island communities. Accordingly, policies should combine targeted economic incentives and regulatory support with accessible technical assistance and community-based implementation mechanisms to facilitate renewable energy uptake in geographically constrained areas.

**Keywords:** Renewable energy adoption; Solar photovoltaic; Island communities; Consumer behavior; Stimulus organism response; Structural equation modeling

## 1. Introduction

The global energy transition toward renewable sources has become increasingly urgent in addressing climate change and energy security challenges. Solar photovoltaic (PV) technology has emerged as a promising solution, particularly for developing countries with abundant solar resources. Electricity generation from renewable sources, such as solar PV, will significantly increase the share of renewable energy in future electricity production. Indonesia, located in the tropical zone with high solar irradiance (approximately 4.8 $kWh \cdot m^{-2} \cdot day^{-1}$), has exceptional potential for solar energy development (Afandi et al., 2022). Despite the Indonesian government's substantial investment in renewable energy infrastructure, household-level adoption of solar PV systems remains well below potential levels. The Indonesia Energy Transition Outlook 2025 emphasizes the need to accelerate renewable energy deployment and strengthen policy and regulatory support for Indonesia's energy transition, while solar PV, including rooftop systems, is expected to play an important role in advancing the country's renewable energy and net-zero emissions ambitions (Institute for Essential Services Reform, 2024a; Institute for Essential Services Reform, 2024b). Indonesia's archipelagic geography creates unique energy challenges, particularly for remote island communities. The Seribu Islands, near Jakarta, face chronic electricity supply constraints, with many settlements relying on expensive diesel generators for power generation. This situation creates both a critical need and market opportunity for distributed solar solutions. While the Indonesian

government has introduced net metering policies and various incentive schemes, adoption rates among island households remain modest. Understanding the specific factors driving household adoption decisions in island contexts is essential for policy formulation and energy market development.

Previous research on renewable energy adoption has primarily focused on developed nations and urban contexts, leaving a significant gap in understanding adoption behavior in developing island regions (Mah et al., 2018). Existing studies emphasize economic factors and technological considerations but provide limited insight into how external stimuli (economic, social, environmental, regulatory, technical, and market factors) interact with psychological variables to shape adoption intentions (Crago & Chernyakhovskiy, 2017). Additionally, few studies have employed integrated theoretical frameworks that combine behavioral economics with technology adoption models in island community contexts. In addition, artificial intelligence-based analyses have been widely reported, including the use of artificial neural network algorithms for various purposes (Familiana et al., 2017; Mamun et al., 2020; Zhang et al., 2022).

The use of stimulus organism response (SOR) as the primary theoretical framework, informed by selected Theory of Planned Behavior (TPB) and Technology Acceptance Model (TAM) concepts, draws on their complementary explanatory roles rather than a simple aggregation of constructs. SOR provides the overarching process architecture by explaining how external conditions (stimuli) influence an individual's internal evaluative state (organism), which subsequently shapes a behavioral response. In this context, economic, environmental, social, regulatory, technical, and market conditions represent contextual stimuli surrounding household solar PV adoption; attitude represents the principal internal evaluative state; and adoption intention represents the behavioral response. However, SOR alone does not specify how technology-related evaluations or behavioral intentions form. TAM addresses this limitation by explaining how perceptions of technological usefulness, ease, complexity, and feasibility contribute to technology evaluation and acceptance (Davis, 1989; Puiu et al., 2025; Siuda & Gręboszk-Krawczyk, 2025). In contrast, TPB provides the behavioral rationale linking favorable evaluation of a behavior, particularly attitude, to behavioral intention (Ajzen, 1991). Other methods, such as structural equation modeling (SEM), can also be applied in the banking sector (Al Mamun et al., 2025).

The global renewable energy sector has experienced unprecedented growth, with renewable sources generating 13.1% of global electricity in 2022. Solar PV technology has been the fastest-growing renewable energy source, with cumulative capacity increasing from 227 GW in 2015 to over 1,200 GW by 2023. In developing nations, renewable energy adoption faces distinct barriers compared to developed economies. Limited access to capital, inadequate grid infrastructure, skills gaps, and weak institutional frameworks constrain widespread adoption. Analysis of sustainable energy transitions in developing countries shows that financial gaps remain a critical constraint despite policy progress (O'Connell & Schot, 2024). For island and remote communities, geographic isolation, higher electricity costs from diesel generators, and limited vendor availability create additional challenges (Rebucas et al., 2024). However, these same factors create strong economic incentives for decentralized renewable energy solutions. Limited research has specifically examined solar adoption in island developing communities, creating a critical research gap that this study addresses. Qureshi et al. (2017) identified cost, technical complexity, and lack of government support as principal barriers to household solar adoption in Pakistan. Evidence from Malaysia similarly indicates that behavioral and social factors contribute to household PV adoption intention (Alam et al., 2021).

In this study, economic conditions, regulatory environments, social influences, environmental concerns, technical characteristics, and market availability represent stimulus categories affecting adoption. Davis (1989) proposed that technology adoption is primarily determined by perceived usefulness and perceived ease of use. These perceptions are shaped by external variables including system characteristics, user characteristics, and implementation processes. Recent 2025 research applying TAM to renewable energy adoption in Poland confirmed the significant positive effects of perceived ease of use and perceived usefulness on purchase intentions for green energy technology (Siuda & Gręboszk-Krawczyk, 2025). Notably, Perceived Ease of Use demonstrated direct positive effects on adoption intentions, even accounting for natural apprehensions regarding learning to use new technologies. Ajzen (1991) posited that behavioral intention is determined by three factors including attitude toward the behavior, subjective norms reflecting social pressure, and perceived behavioral control regarding ability to execute the behavior. TPB has demonstrated robust predictive validity across numerous behavior domains including technology adoption. Extended TPB research demonstrates that integrating organizational factors with core TPB constructs substantially enhances predictive validity for energy-related behaviors, with perceived behavioral control emerging as a particularly strong predictor of energy-saving intentions (Ajzen, 1991; Puiu et al., 2025). Within this SOR-based framework, external stimuli are treated as drivers of attitude formation. Attitude, in turn, represents the organismic factor determining adoption intention and actual behavior. This integration aligns with value-attitude-behavior hierarchies documented in consumer behavior literature and reflects recent advances in multi-theory adoption research.

Against this background, this study addresses three research questions: (RQ1) How do economic, environmental, social, regulatory, technical, and market stimuli influence household attitudes toward solar PV and their adoption intentions? (RQ2) To what extent does attitude toward solar PV mediate the relationships between these external

stimuli and adoption intention? and (RQ3) Which determinants exert the strongest influence on household solar PV adoption intention in an island-community context? Accordingly, the study aims to identify and quantify the effects of the six external stimuli on attitude and adoption intention, examine attitude's mediating role in translating external conditions into behavioral intention, and determine the most influential drivers and barriers shaping household solar PV adoption intention.

To address these objectives, we conducted a cross-sectional household survey in Pulau Tidung, Seribu Islands, Indonesia. Data from 100 household heads or primary energy decision-makers, selected through purposive sampling, were analyzed using partial least squares structural equation modeling (PLS-SEM). The analysis evaluates both the measurement and structural models and estimates direct and indirect relationships among external stimuli, attitude, and solar PV adoption intention using bootstrapping.

This study makes three principal contributions. First, theoretically, the study develops an SOR-based explanatory framework in which external contextual stimuli shape internal psychological evaluation and subsequent adoption intention, while selected TPB and TAM concepts provide complementary interpretation of behavioral and technology-related mechanisms. Second, empirically, it extends the renewable energy adoption literature by providing evidence from a developing-country island community, a context that remains comparatively underrepresented in household solar PV research. Third, from a practical perspective, it distinguishes the economic, regulatory, and technical mechanisms associated with household adoption intention, thereby providing evidence for designing context-sensitive incentives, regulatory support, technical assistance, and community-based implementation strategies. These contributions are particularly relevant to geographically constrained communities where infrastructure, market accessibility, and technical capacity can substantially influence the diffusion of distributed renewable energy technologies.

Within the SOR framework, external economic, environmental, and social conditions can shape the internal evaluation households use to assess solar PV adoption. Economic considerations are particularly relevant because the technology involves substantial initial investment but may generate long-term electricity-cost savings. Previous research has identified financial incentives, affordability, and household economic considerations as important determinants of residential solar PV adoption (Crago & Chernyakhovskiy, 2017; Mundaca & Samahita, 2020). Favorable economic conditions should therefore strengthen households' evaluation of solar PV.

H1: Economic stimulus positively influences household attitude toward solar PV adoption.

Environmental concern may similarly strengthen favorable attitudes because solar PV is associated with cleaner energy production and reduced environmental impacts. Previous research on public attitudes toward PV development has demonstrated the relevance of environmental considerations in shaping acceptance of solar technologies (Tsantopoulos et al., 2014).

H2: Environmental stimulus positively influences household attitude toward solar PV adoption.

Social interaction provides another stimulus through family recommendations, community expectations, peer experience, and technology visibility. Prior studies have documented the importance of peer and social influences in residential renewable energy adoption (Alam et al., 2021; Graziano et al., 2019).

H3: Social stimulus positively influences household attitude toward solar PV adoption.

Some external conditions may operate more directly on behavioral intention by facilitating or constraining households' ability and willingness to adopt solar PV. Regulatory support can reduce institutional uncertainty through stable policies, incentives, and government commitment, and previous studies show that policy incentives and institutional support can influence residential PV adoption decisions (Crago & Chernyakhovskiy, 2017; Mah et al., 2018).

H4: Regulatory stimulus positively influences household solar PV adoption intention.

In contrast, technical complexity, maintenance requirements, reliability concerns, and the need for specialized expertise may discourage adoption. This expectation aligns with TAM, in which ease-related technology perceptions influence acceptance, and with recent renewable-energy applications showing the importance of perceived ease of use for adoption intention (Davis, 1989; Siuda & Gręboszk-Krawczyk, 2025).

H5: Technical stimulus, conceptualized as perceived technical barriers, negatively influences household solar PV adoption intention.

Market conditions may also determine whether households can readily access products, suppliers, information, installation services, and after-sales support. Previous research shows that market imperfections and accessibility constraints can influence investment in stand-alone solar PV systems (Abdul-Salam & Phimister, 2019), and limited market and infrastructure access has also been identified as relevant to potential adopters (Mah et al., 2018). This consideration is especially pertinent to geographically isolated island communities.

H6: Market stimulus positively influences household solar PV adoption intention.

Finally, attitude represents the internal evaluative mechanism linking external conditions with behavioral response. TPB identifies attitude as a key antecedent of behavioral intention (Ajzen, 1991), while TAM similarly emphasizes the role of favorable technology evaluation in acceptance decisions (Davis, 1989). Empirical research on household PV adoption has also demonstrated the importance of behavioral and attitudinal factors in shaping intention (Alam et al., 2021).

H7: Attitude toward solar PV positively influences household adoption intention.

Consistent with the SOR mechanism, we examine the mediating role of attitude between external stimuli and adoption intention separately through bootstrapped indirect-effect analysis and do not designate it as an additional numbered hypothesis. Selected TPB- and TAM-related constructs, including subjective norm and perceived behavioral control, are represented as lower-order measurement components within the hierarchical specification rather than as independent structural predictors. Their role is therefore measurement-related and should not be interpreted as additional canonical TPB structural paths.

In addition to the seven a priori hypotheses, the structural model estimated supplementary exploratory paths from each contextual stimulus to the alternative endogenous construct. Specifically, technical, regulatory, and market stimuli were additionally estimated as predictors of attitude, whereas economic, environmental, and social stimuli were additionally estimated as predictors of adoption intention. These paths were not treated as confirmatory hypotheses or control paths; rather, they were included as supplementary exploratory relationships to examine whether the contextual stimuli exhibited effects beyond their theoretically hypothesized pathways.

## 2. Research Methodology

### 2.1 Research Design and Setting

This study employs a cross-sectional survey design implemented in Pulau Tidung (Tidung Island), part of the Seribu Islands archipelago, located approximately 15 kilometers north of Jakarta. Pulau Tidung represents an ideal research setting as it exhibits characteristics typical of remote island communities while remaining accessible for data collection. The island's population of approximately 3,200 individuals relies partially on expensive diesel generators for electricity. Approximately 89% of surveyed households expressed willingness to adopt solar PV systems, indicating substantial market potential despite actual adoption rates below 5%.

### 2.2 Sampling and Participants

The research employed non-probability purposive sampling, targeting household heads or primary decision-makers responsible for energy consumption decisions. This sampling approach was selected based on research context specificity, as the study focused on understanding adoption behavior among relevant decision-makers rather than generating population-representative estimates. Sample size ($n$) determination utilized the Lemeshow formula for unknown population parameters using Eq. (1).

$$n = \frac{Z^2 \times P(1-P)}{e^2} \tag{1}$$

where, $Z$ = 1.96 (95% confidence level), $P$ = 0.5 (conservative maximum variance estimate), and $e$ = 0.10 (acceptable error margin). The final sample comprised 100 households. Detailed respondent characteristics, including frequencies, percentages, and valid denominators, are reported in Section 3.1. We used the Lemeshow calculation as a pragmatic minimum recruitment target for the household survey, rather than as a basis for population-representative inference, given the purposive sampling design. To further assess whether the achieved sample was adequate for PLS-SEM, we conducted a retrospective sensitivity analysis using the inverse square root approach proposed by Kock & Hadaya (2018). At a two-tailed significance level of 0.05 and 80% statistical power, a sample of 100 corresponds to a minimum detectable standardized path coefficient of approximately 0.249. The absolute magnitudes of the statistically significant structural path coefficients observed in the present model ranged from $|\beta|$ = 0.306 to 0.827 and therefore exceeded this sensitivity threshold. This assessment supports the sample's adequacy for detecting the moderate-to-large relationships identified in the study, while recognizing that weaker effects may require larger samples for reliable detection.

### 2.3 Data Collection

Data collection occurred between July and August 2025 through multiple modalities combining in-person surveys and online questionnaires to accommodate geographic constraints. Trained enumerators administered questionnaires with respondent assistance. The research team coordinated with local government officials and community leaders to facilitate respondent access and enhance data quality through preliminary educational sessions explaining solar technology concepts. Participation was voluntary. Before questionnaire administration, respondents were verbally informed about the purpose of the study, the nature of their participation, the confidentiality of their responses, and their right to decline participation. Verbal informed consent was obtained from each respondent before proceeding with the survey. Survey instruments employed 5-point Likert scales (strongly disagree = 1 to strongly agree = 5) for all measurement items. Questionnaire items were adapted from

established solar PV adoption and behavioral-intention studies (Alam et al., 2021; Tsantopoulos et al., 2014), while the attitude and behavioral-intention measures were theoretically grounded in TAM and TPB (Ajzen, 1991; Davis, 1989). Before the main survey, the draft questionnaire underwent a cognitive pretest with 10 respondents drawn from the target population to assess item comprehension, wording clarity, contextual relevance, and interpretation of the response options. Feedback from the cognitive pretest was used to refine ambiguous or contextually inappropriate wording before final questionnaire administration. This cognitive pretest was intended to assess comprehensibility and contextual suitability and was not treated as a formal psychometric reliability pilot. Data quality procedures included checking for missing values and outliers, assessing measurement model reliability and validity, evaluating potential common method bias using Harman's single-factor test and a full-collinearity variance inflation factors (VIFs) assessment, and examining the distributional characteristics of the data. For the descriptive analysis, frequencies and percentages were calculated using the valid non-missing responses for each variable. Where demographic information was missing, the corresponding valid denominator is reported explicitly in the Results. No survey weighting was applied. The final study sample comprised 100 households.

### 2.4 Measurement Scales

The final SmartPLS specification employed a hierarchical measurement structure comprising six contextual stimulus constructs, two higher-order focal constructs, and nine lower-order behavioral and technology-related components. Each contextual stimulus—economic, environmental, social, regulatory, technical, and market—was measured using four indicators, resulting in 24 contextual-stimulus indicators. Attitude was specified as a higher-order construct represented by five lower-order components (affective attitude (AA), behavioral intention toward continued use (BI), perceived ease of use (PEOU), perceived usefulness (PU), and technology anxiety (TA)), each measured using five indicators. Adoption intention was specified as a higher-order construct represented by four lower-order components (attitude toward adoption (ATA), behavioral intention toward initial adoption (BN), perceived behavioral control (PBC), and subjective norm (SN)), each measured using four indicators. Thus, the final empirical specification contained 65 unique observed indicators. Economic stimulus assessed perceived economic benefits, including the affordability of initial investment, long-term cost-saving potential, availability of financing options, and reduced monthly electricity expenditure. Environmental stimulus measured ecological consciousness, including concern about climate change, perceived environmental damage from current energy sources, desire to reduce carbon footprint, and belief in the importance of sustainable development. Social stimulus captured social influences, including family recommendations, community norms toward renewable energy, peer influence from adopters, and perceived social acceptability of solar adoption. Regulatory stimulus evaluated perceived government support through awareness of renewable energy policies, perceived adequacy of government incentives, expectations regarding policy stability, and perceived policy supportiveness. Technical stimulus assessed perceived technical barriers, including installation complexity, maintenance difficulty and cost, technology reliability and durability, and the perceived need for technical expertise. Market stimulus assessed respondents' perceptions of market accessibility, including the availability of solar PV products, suppliers, information, installation services, and after-sales support. At the higher-order level, attitude and adoption intention were modeled as hierarchical constructs in the SmartPLS specification. Attitude was represented by five lower-order components coded AA, BI, PEOU, PU, and TA, whereas adoption intention was represented by four lower-order components coded ATA, BN, PBC, and SN. These lower-order components form part of the measurement structure of the two higher-order constructs and were not treated as additional independent structural predictors. Accordingly, the structural hypotheses of the study concern the relationships among the six contextual stimuli, attitude, and adoption intention, while the lower-order components are used to operationalize the hierarchical measurement structure. The psychometric properties of the final measurement model, including internal consistency reliability, composite reliability, convergent validity, and discriminant validity, were evaluated using the full analytical sample.

### 2.5 Data Analysis

Data analysis employed a two-stage PLS-SEM approach following established methodological recommendations (Hair et al., 2013; Hair et al., 2022; Sarstedt et al., 2022). PLS-SEM was selected because the study focuses on estimating a multi-construct explanatory structural model involving direct and indirect relationships. In Stage 1, the measurement model was evaluated using indicator loadings, Cronbach's $\alpha$, Dijkstra–Henseler reliability coefficient (rho_A), composite reliability (rho_C), average variance extracted (AVE), the Fornell–Larcker criterion, cross-loadings, and the heterotrait–monotrait ratio (HTMT). In Stage 2, the structural model was assessed using path coefficients, inner-model VIF, $R^2$ and adjusted $R^2$ values, and effect sizes ($f^2$). In addition to the structural-model VIF assessment, full-collinearity VIFs were examined across the focal latent constructs as a supplementary diagnostic for potential common method bias and pervasive construct collinearity. Following Kock (2015), a value of 3.3 was used as a conservative diagnostic benchmark. Statistical significance

was evaluated through bootstrapping with 5,000 resamples, using two-tailed tests at $\alpha = 0.05$, together with bootstrap confidence intervals (CIs). Direct and indirect effects were evaluated separately. SmartPLS 4.0 was used for all analyses. Because no separate out-of-sample predictive assessment was conducted, the model is interpreted primarily in explanatory terms, and no claim of out-of-sample predictive performance is made.

## 3. Results and Discussion

### 3.1 Descriptive Statistics and Sample Characteristics

The final study sample comprised 100 households. Descriptive results are reported as percentages (%). Percentages were calculated using the valid non-missing denominator for each variable, and no survey weighting was applied. The 100 surveyed households exhibited diverse characteristics reflecting typical island community demographics. Male respondents constituted 82.2% of the sample, reflecting the tradition of male household heads managing major infrastructure decisions. Age distribution showed 58.4% aged 30 years or older, representing the household decision-making demographic. Educational attainment included 87% completing secondary education or higher, suggesting reasonable capacity for understanding technical information regarding solar systems. Occupational distribution indicated 41.8% self-employed (primarily fishing and tourism-related enterprises) and 33.7% currently unemployed or irregularly employed. Monthly household income for 77.5% of respondents fell between Rp 1–3 million (USD 65–195), typical for remote island communities. Monthly electricity expenditure ranged from Rp 100,000–300,000 (USD 6.50–19.50) for 60.3% of respondents, reflecting reliance on expensive diesel generators. Importantly, 89.1% of respondents expressed willingness to adopt solar PV systems, and 92.1% supported community-based (cooperative) management models. These figures suggest substantial latent demand for solar solutions if adoption barriers can be addressed.

### 3.2 Measurement and Structural Model Results

Using the full analytical sample ($n = 100$), the final hierarchical measurement model was evaluated for indicator reliability, internal consistency reliability, and convergent validity. The SmartPLS specification comprises six first-order contextual stimulus constructs, two higher-order focal constructs, attitude and adoption intention, and their corresponding lower-order components. Attitude is represented by five lower-order components (AA, BI, PEOU, PU, and TA), whereas adoption intention is represented by four lower-order components (ATA, BN, PBC, and SN). Table 1 summarizes the measurement-model statistics across these contextual, lower-order, and higher-order constructs, including indicator-loading ranges, Cronbach's $\alpha$, rho_A, rho_C, and AVE. For the higher-order constructs, the reported loading ranges reflect the repeated indicators associated with their respective lower-order components, whereas the loading ranges for the contextual and lower-order constructs represent their direct indicator loadings.

As shown in Table 1, all reported indicator loadings exceeded the conventional 0.70 benchmark, with loading ranges across the hierarchical measurement model extending from 0.865 to 0.975. Cronbach's $\alpha$ values ranged from 0.935 to 0.994, rho_A from 0.936 to 0.994, and rho_C from 0.954 to 0.994. AVE values ranged from 0.837 to 0.937, exceeding the recommended minimum of 0.50 and therefore supporting convergent validity across the reported constructs and components. These results indicate strong indicator reliability and internal consistency within the hierarchical measurement specification. Nevertheless, several reliability coefficients, particularly for the higher-order attitude and adoption intention constructs and some of their lower-order components, were exceptionally high. Such values may reflect substantial within-construct homogeneity or indicator redundancy and are therefore interpreted cautiously rather than as evidence of superior measurement quality alone.

Discriminant validity was examined using the Fornell–Larcker criterion, cross-loadings, and the HTMT. Table 2 presents the HTMT matrix for the focal constructs. Inspection of the combined discriminant-validity diagnostics indicated that construct distinctiveness was not uniformly established across all construct pairs. Several HTMT values exceeded the conventional 0.90 threshold, particularly among several contextual stimuli. Because subjective norm is modeled as a lower-order component of adoption intention, the high association between these two constructs is not interpreted as an independent discriminant-validity failure. Accordingly, discriminant validity is interpreted cautiously rather than considered uniformly established, and structural relationships involving closely related constructs should be interpreted with corresponding caution.

Table 3 provides the construct correlation matrix together with the Fornell–Larcker criterion, where the diagonal elements represent the square roots of AVE and the off-diagonal elements represent inter-construct correlations. The results indicate that discriminant validity was not uniformly established across all construct pairs. In several cases, the inter-construct correlations exceeded the corresponding square roots of AVE. For example, the correlation between regulatory and technical stimuli (0.954) exceeded the square roots of AVE for both regulatory stimulus (0.952) and technical stimulus (0.950). Similarly, the regulatory–market correlation (0.953) exceeded the corresponding diagonal values of 0.952 and 0.948. High correlations were also observed among several contextual

stimuli, including economic, regulatory, market, social, and technical constructs. These results are consistent with the elevated HTMT values reported in Table 2. Construct correlations and Fornell–Larcker criterion indicate limited discriminant separation among several conceptually related constructs. Accordingly, the structural relationships involving these highly correlated constructs should be interpreted cautiously.

**Table 1.** Measurement model assessment of the final hierarchical SmartPLS constructs

| Construct | Model Level | Components | Loading Range | Cronbach's $\alpha$ | rho_A | rho_C | AVE |
|---|---|---|---|---|---|---|---|
| Economic stimulus | First-order contextual | 4 indicators | 0.933–0.961 | 0.963 | 0.964 | 0.973 | 0.901 |
| Environmental stimulus | First-order contextual | 4 indicators | 0.906–0.941 | 0.942 | 0.945 | 0.958 | 0.852 |
| Social stimulus | First-order contextual | 4 indicators | 0.865–0.958 | 0.935 | 0.936 | 0.954 | 0.837 |
| Regulatory stimulus | First-order contextual | 4 indicators | 0.943–0.964 | 0.966 | 0.966 | 0.975 | 0.906 |
| Technical stimulus | First-order contextual | 4 indicators | 0.942–0.958 | 0.964 | 0.965 | 0.974 | 0.903 |
| Market stimulus | First-order contextual | 4 indicators | 0.937–0.953 | 0.962 | 0.963 | 0.973 | 0.899 |
| Attitude | Higher-order | 5 LOCs/25 repeated indicators | 0.895–0.956 | 0.994 | 0.994 | 0.994 | 0.871 |
| Adoption intention | Higher-order | 4 LOCs/16 repeated indicators | 0.879–0.963 | 0.990 | 0.990 | 0.990 | 0.865 |
| AA | Lower-order of attitude | 5 indicators | 0.940–0.955 | 0.971 | 0.971 | 0.977 | 0.895 |
| BI | Lower-order of attitude | 5 indicators | 0.949–0.963 | 0.978 | 0.979 | 0.983 | 0.921 |
| PEOU | Lower-order of attitude | 5 indicators | 0.921–0.962 | 0.971 | 0.972 | 0.978 | 0.898 |
| PU | Lower-order of attitude | 5 indicators | 0.944–0.970 | 0.975 | 0.975 | 0.98 | 0.909 |
| TA | Lower-order of attitude | 5 indicators | 0.942–0.965 | 0.973 | 0.973 | 0.979 | 0.904 |
| ATA | Lower-order of adoption intention | 4 indicators | 0.938–0.967 | 0.969 | 0.970 | 0.977 | 0.915 |
| BN | Lower-order of adoption intention | 4 indicators | 0.924–0.948 | 0.955 | 0.956 | 0.967 | 0.881 |
| PBC | Lower-order of adoption intention | 4 indicators | 0.947–0.965 | 0.970 | 0.971 | 0.978 | 0.918 |
| SN | Lower-order of adoption intention | 4 indicators | 0.950–0.975 | 0.978 | 0.978 | 0.984 | 0.937 |

Note: rho_A = Dijkstra–Henseler reliability coefficient; rho_C = composite reliability; AVE = average variance extracted; LOC = lower-order component; AA = affective attitude; BI = behavioral intention toward continued use; PEOU = perceived ease of use; PU = perceived usefulness; TA = technology anxiety; BN = behavioral intention toward initial adoption; ATA = attitude toward adoption; PBC = perceived behavioral control; SN = subjective norm.

**Table 2.** HTMT matrix

| Construct | Attitude | Adoption Intention | IE | IEC | IM | IR | IS | IT | SN |
|---|---|---|---|---|---|---|---|---|---|
| **Attitude** | | | | | | | | | |
| **Adoption Intention** | 0.975 | | | | | | | | |
| **IE** | 0.862 | 0.87 | | | | | | | |
| **IEC** | 0.923 | 0.923 | 0.924 | | | | | | |
| **IM** | 0.896 | 0.888 | 0.912 | 0.966 | | | | | |
| **IR** | 0.913 | 0.928 | 0.922 | 0.988 | 0.988 | | | | |
| **IS** | 0.908 | 0.928 | 0.926 | 0.976 | 0.96 | 0.999 | | | |
| **IT** | 0.918 | 0.9 | 0.929 | 0.972 | 0.979 | 0.989 | 0.984 | | |
| **SN** | 0.959 | 0.998 | 0.821 | 0.894 | 0.845 | 0.9 | 0.903 | 0.862 | |

Note: HTMT = heterotrait–monotrait ratio; IE = environmental stimulus; IEC = economic stimulus; IM = market stimulus; IR = regulatory stimulus; IS = social stimulus; IT = technical stimulus; SN = subjective norms.

Harman's single-factor test showed that the largest single factor accounted for 38% of the total variance, below

the conventional 50% threshold. However, because this diagnostic alone is insufficient to rule out common method bias, it was supplemented with the full-collinearity VIF assessment reported in Table 4. As shown in Table 4, the full-collinearity VIF values ranged from 5.748 for environmental stimulus to 59.042 for adoption intention, with all focal constructs exceeding the conservative threshold of 3.3. Particularly high values were observed for adoption intention (VIF = 59.042), subjective norms (VIF = 35.144), regulatory stimulus (VIF = 26.749), and attitude (VIF = 21.576). These results indicate substantial multivariate collinearity among the focal constructs and suggest that potential common method bias cannot be ruled out. However, elevated full-collinearity VIF values should not be interpreted as definitive evidence that common method bias alone generated the observed relationships, because substantive conceptual overlap among highly related constructs may also contribute to the elevated values. Therefore, the structural coefficients are interpreted cautiously, and the potential influence of common method bias and construct collinearity is explicitly acknowledged as a limitation of the study.

**Table 3.** Construct correlations and Fornell–Larcker criterion

| Construct | Attitude | Adoption Intention | IE | IEC | IM | IR | IS | IT | SN |
|---|---|---|---|---|---|---|---|---|---|
| **Attitude** | 0.933 | | | | | | | | |
| **Adoption Intention** | 0.967 | 0.930 | | | | | | | |
| **IE** | 0.836 | 0.841 | 0.923 | | | | | | |
| **IEC** | 0.903 | 0.902 | 0.882 | 0.949 | | | | | |
| **IM** | 0.877 | 0.866 | 0.870 | 0.930 | 0.948 | | | | |
| **IR** | 0.894 | 0.907 | 0.881 | 0.953 | 0.953 | 0.952 | | | |
| **IS** | 0.876 | 0.892 | 0.869 | 0.927 | 0.911 | 0.949 | 0.915 | | |
| **IT** | 0.899 | 0.879 | 0.886 | 0.937 | 0.943 | 0.954 | 0.935 | 0.950 | |
| **SN** | 0.946 | 0.982 | 0.791 | 0.868 | 0.82 | 0.875 | 0.864 | 0.838 | 0.968 |

Note: Diagonal elements in bold are the square roots of AVE; off-diagonal elements are construct correlations. IE = environmental stimulus; IEC = economic stimulus; IM = market stimulus; IR = regulatory stimulus; IS = social stimulus; IT = technical stimulus; SN = subjective norms.

**Table 4.** Full-collinearity VIF

| Construct | Full-Collinearity VIF |
|---|---|
| Attitude | 21.576 |
| Adoption intention | 59.042 |
| Environmental stimulus | 5.748 |
| Economic stimulus | 14.189 |
| Market stimulus | 13.988 |
| Regulatory stimulus | 26.749 |
| Social stimulus | 12.405 |
| Technical stimulus | 18.204 |
| Subjective norms | 35.144 |

Note: VIF = variance inflation factors. Full-collinearity VIF values were calculated from the focal-construct latent-variable correlation matrix. Following the conservative 3.3 benchmark, all reported values exceed the threshold and therefore indicate that potential common method bias and/or pervasive construct collinearity cannot be ruled out.

Figure 1 presents the hierarchical PLS-SEM specification actually used in the SmartPLS analysis. The six contextual stimuli, economic, environmental, social, regulatory, technical, and market, are modeled as predictors of the higher-order attitude and adoption-intention constructs. Attitude is additionally linked to adoption intention, representing the central organism-to-response pathway in the SOR-based framework. The additional latent variables displayed around the two higher-order constructs are lower-order measurement components rather than additional independent structural predictors: AA, BI, PEOU, PU, and TA form the hierarchical measurement specification of attitude, whereas ATA, BN, PBC, and SN form the hierarchical measurement specification of adoption intention. Accordingly, the larger number of latent variables and indicators visible in Figure 1 reflects the hierarchical measurement structure used in SmartPLS rather than a different structural model. The focal structural relationships among the contextual stimuli, attitude, and adoption intention are reported in Table 5.

The structural model was subsequently evaluated using path coefficients, bootstrapped significance tests, bias-corrected CI, effect sizes ($f^2$), and inner-model VIF values. Table 5 summarizes the direct structural relationships. The measurement relationships between the higher-order constructs and their lower-order components shown in Figure 1 are not treated as additional structural hypotheses and are therefore not reproduced as structural paths in Table 5. The hypothesized paths are evaluated against H1–H7, whereas the additional cross-outcome paths are reported separately as supplementary exploratory relationships and are not used to redefine the original hypotheses. The principal hypothesis decisions remain consistent with the previously reported results: H1, H4, H5, and H7 were supported, whereas H2, H3, and H6 were not supported at the prespecified $\alpha = 0.05$ level. Table 5 presents comprehensive direct effect results. Attitude toward solar PV had an exceptionally strong positive effect on

adoption intention ($\beta$ = 0.827, $t$ = 8.957, $p$ < 0.001), strongly supporting H7. This relationship's magnitude indicates that attitude represents a primary psychological mechanism through which other factors influence adoption decisions, consistent with recent meta-analyses validating the central role of attitude in behavioral intention models (Leong et al., 2023). Economic stimulus significantly and positively influenced attitude ($\beta$ = 0.411, $t$ = 2.376, $p$ = 0.018), supporting H1. This finding suggests that household economic considerations, including purchase affordability, long-term savings potential, and financing availability, may contribute to more favorable attitudes toward solar adoption. However, economic stimulus showed negligible direct effect on adoption intention ($\beta$ = 0.000, $p$ = 0.999), indicating that economic influences operate primarily through attitude formation rather than directly determining behavioral intention.

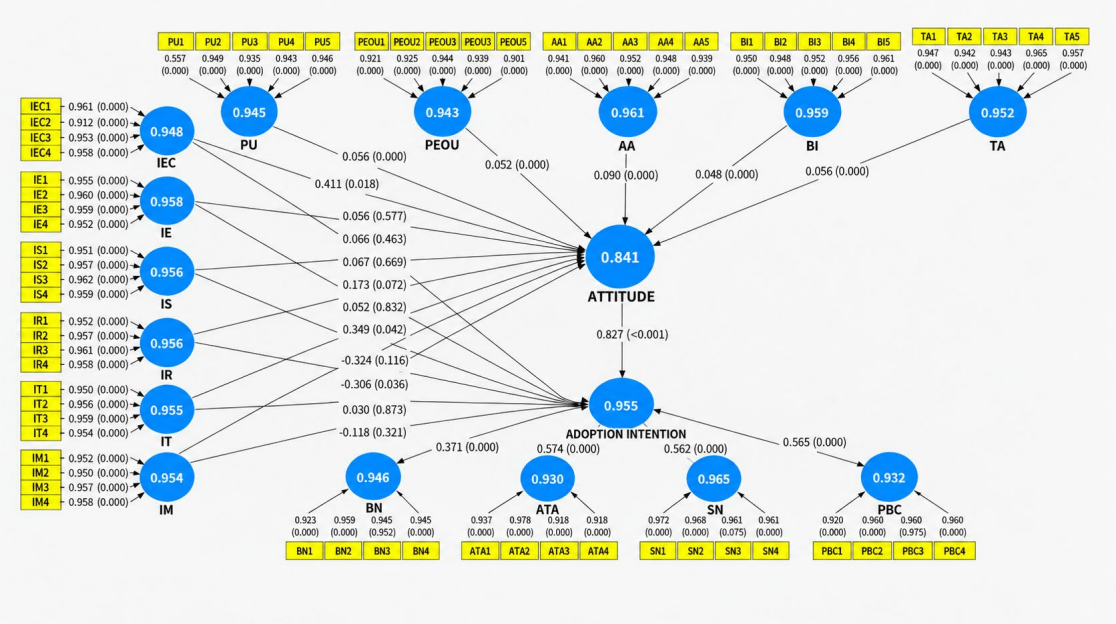



**Figure 1.** Hierarchical PLS-SEM specification showing the contextual stimuli, higher-order attitude and adoption intention constructs, their lower-order components, and the estimated structural relationships

Note: PLS-SEM = partial least squares structural equation modeling; AA = affective attitude; BI = behavioral intention toward continued use; PEOU = perceived ease of use; PU = perceived usefulness; TA = technology anxiety; BN = behavioral intention toward initial adoption; ATA = attitude toward adoption; PBC = perceived behavioral control; SN = subjective norm; IE, IS, IR, IT, IM, and IEC represent environmental, social, regulatory, technical, market, and economic stimuli, respectively.

**Table 5.** Structural model assessment

| Relationship | $\beta$ | $t$ | $p$ | Bias-Corrected 95% CI | $f^2$ | Inner VIF |
|---|---|---|---|---|---|---|
| Economic → Attitude | 0.411 | 2.376 | 0.018 | [0.069, 0.745] | 0.081 | 13.127 |
| Environmental → Attitude | 0.056 | 0.558 | 0.577 | [−0.124, 0.284] | 0.004 | 5.324 |
| Social → Attitude | 0.067 | 0.428 | 0.669 | [−0.168, 0.458] | 0.002 | 11.632 |
| Technical → Attitude | 0.324 | 1.571 | 0.116 | [−0.078, 0.727] | 0.043 | 15.214 |
| Regulatory → Attitude | 0.052 | 0.212 | 0.832 | [−0.417, 0.559] | 0.001 | 23.474 |
| Market → Attitude | 0.030 | 0.159 | 0.873 | [−0.388, 0.354] | <0.001 | 13.055 |
| Attitude → Adoption intention | 0.827 | 8.957 | <0.001 | [0.644, 1.017] | 2.412 | 6.279 |
| Economic → Adoption intention | 0 | 0.001 | 0.999 | [−0.313, 0.302] | <0.001 | 14.188 |
| Environmental → Adoption intention | 0.066 | 0.734 | 0.463 | [−0.060, 0.303] | 0.018 | 5.343 |
| Social → Adoption intention | 0.173 | 1.801 | 0.072 | [0.010, 0.391] | 0.057 | 11.660 |
| Technical → Adoption intention | −0.306 | 2.100 | 0.036 | [−0.564, −0.011] | 0.131 | 15.874 |
| Regulatory → Adoption intention | 0.349 | 2.037 | 0.042 | [0.052, 0.742] | 0.115 | 23.491 |
| Market → Adoption intention | −0.118 | 0.993 | 0.321 | [−0.358, 0.120] | 0.024 | 13.060 |

Note: VIF = variance inflation factors; CI = confidence interval. Statistical decisions were based on the prespecified two-tailed significance threshold of $p$ < 0.05. Accordingly, the social stimulus → adoption intention path was treated as non-significant ($p$ = 0.072), although its bias-corrected bootstrap confidence interval did not include zero.

Table 5 indicates that the principal substantive pattern remains broadly unchanged after the expanded structural-model assessment. Attitude showed the largest positive association with adoption intention ($\beta$ = 0.827, $p$ < 0.001) and was associated with the largest observed effect size ($f^2$ = 2.412). Economic stimulus significantly influenced attitude ($\beta$ = 0.411, $p$ = 0.018; $f^2$ = 0.081), whereas regulatory stimulus positively affected adoption intention ($\beta$ =

0.349, $p = 0.042$; $f^2 = 0.115$) and technical stimulus exerted a significant negative effect ($\beta = -0.306$, $p = 0.036$; $f^2 = 0.131$). The remaining hypothesized effects were not statistically significant at $\alpha = 0.05$. Thus, the additional effect-size information reinforces the central role of attitude while showing that the significant contextual effects were comparatively smaller.

The inner-model VIF values ranged from 5.324 to 23.491 for the contextual predictors, while the VIF for attitude as a predictor of adoption intention was 6.279. These values indicate substantial collinearity among several predictors in the current model specification. The individual path coefficients are therefore interpreted cautiously, particularly for contextual variables entered simultaneously into the same endogenous equation. This diagnostic does not change the reported coefficient estimates or hypothesis decisions, but it limits the extent to which closely related predictor effects can be interpreted as fully independent contributions.

Technical barriers had a statistically significant negative effect on adoption intention ($\beta = -0.306$, $t = 2.100$, $p = 0.036$), consistent with TAM-related expectations and supporting H5. This finding indicates that perceived installation complexity, maintenance cost concerns, and reliability questions substantially reduce adoption intentions, independent of overall attitude. This negative direct effect suggests technical barriers function partly through psychological pathways distinct from attitude formation. Recent research emphasizes that perceived ease of use, a key TAM construct, remains critical for predicting adoption behavior even among technology-favorable populations (Siuda & Gręboszk-Krawczyk, 2025). Regulatory stimulus significantly and directly enhanced adoption intention ($\beta = 0.349$, $t = 2.037$, $p = 0.042$), supporting H4. The results suggest that government policies, incentive programs, and perceived institutional support for renewable energy adoption may be positively associated with behavioral intention beyond the contribution of attitude. This finding emphasizes that institutional legitimation provides independent motivation for adoption decisions. Environmental stimulus did not significantly influence attitude ($\beta = 0.056$, $p = 0.577$), and H2 was therefore not supported. Social stimulus likewise showed no significant effect on attitude ($\beta = 0.067$, $p = 0.669$), providing no support for H3. Market stimulus did not significantly influence adoption intention ($\beta = -0.118$, $p = 0.321$), and H6 was therefore not supported. The supplementary direct path from social stimulus to adoption intention was also non-significant at the prespecified $\alpha = 0.05$ level ($\beta = 0.173$, $p = 0.072$). Overall, the results indicate that environmental, social, and market factors did not exert statistically significant effects within the corresponding hypothesized relationships in the present sample.

The model explained 84.1% of the variance in attitude ($R^2 = 0.841$; adjusted $R^2 = 0.831$) and 95.5% of the variance in adoption intention ($R^2 = 0.955$; adjusted $R^2 = 0.951$). These values indicate strong in-sample explanatory power within the present dataset. The particularly high explained variance in adoption intention is partly attributable to the strong proximal relationship between attitude and adoption intention together with the simultaneous inclusion of multiple contextual predictors. Accordingly, the $R^2$ values are interpreted as measures of explanatory performance within the present sample rather than evidence of equivalent out-of-sample predictive accuracy. Because a separate out-of-sample predictive assessment was not conducted, no claim of out-of-sample predictive performance is made.

Mediation analysis examined whether attitude mediated the relationships between the six contextual stimuli and adoption intention using bootstrapped specific indirect effects. Among the six stimuli, only economic stimulus showed a statistically significant indirect effect through attitude (indirect effect = 0.340, $p = 0.032$, bias-corrected 95% CI [0.050, 0.657]), indicating that attitude significantly mediates the relationship between economic stimulus and adoption intention. The corresponding indirect effects were not statistically significant for environmental stimulus ($\beta = 0.046$, $p = 0.587$, 95% CI [−0.100, 0.246]), social stimulus ($\beta = 0.055$, $p = 0.667$, 95% CI [−0.137, 0.375]), regulatory stimulus ($\beta = 0.043$, $p = 0.838$, 95% CI [−0.324, 0.511]), technical stimulus ($\beta = 0.268$, $p = 0.101$, 95% CI [−0.063, 0.578]), or market stimulus ($\beta = 0.024$, $p = 0.875$, 95% CI [−0.343, 0.283]), because their CIs included zero. Thus, the mediation results indicate that attitude primarily transmits the effect of economic conditions to adoption intention, whereas the significant negative effect of technical barriers operates predominantly through a direct pathway rather than through attitude.

### 3.3 Theoretical Implications

The findings primarily support an SOR-based explanatory structure complemented by selected TPB and TAM concepts. SOR provides the overarching distinction between contextual stimuli, internal evaluation, and behavioral response. Within this structure, the significant relationship between economic stimulus and attitude illustrates the stimulus-to-organism mechanism, whereas the strong relationship between attitude and adoption intention reflects the organism-to-response mechanism. TPB provides complementary conceptual grounding for the behavioral constructs included in the model, while TAM informs the interpretation of technology-related complexity, feasibility, and ease-related barriers. Importantly, the empirical model does not reproduce the complete canonical causal structures of TPB or TAM. The theoretical contribution should therefore be interpreted as an SOR-based explanation enriched by selected behavioral and technology-acceptance concepts rather than as a full empirical test of all components of TPB and TAM. SOR provides the overarching mechanism by distinguishing external contextual conditions from internal evaluation and subsequent behavioral response. Within this structure, the

significant effect of economic stimulus on attitude illustrates the stimulus-to-organism mechanism. In contrast, the strong relationship between attitude and adoption intention supports the organism-to-response pathway. TAM adds technology-specific explanatory value by clarifying why perceived technical complexity and maintenance-related concerns can directly inhibit adoption intention. In contrast, TPB provides the behavioral rationale for the strong translation of favorable attitude into intention. This complementary theoretical perspective is therefore informative because the observed effects operate through different pathways. Some contextual stimuli primarily shape internal evaluation, while others more directly facilitate or constrain behavioral intention. This differentiated pattern would be harder to explain by treating SOR, TAM, or TPB as a standalone framework. Second, the results demonstrate that attitude functions as a critical theoretical construct in adoption models. The path coefficient from attitude to adoption intention (0.827) substantially exceeds paths from individual stimuli, confirming that adoption decisions reflect more than simple rational responses to incentives or barriers. Instead, individuals' overall psychological orientations toward solar technology represent a crucial synthesizing mechanism integrating diverse information and influences into coherent behavioral intentions.

Third, the differential effects of various stimuli types reveal that adoption determinants operate through distinct psychological and behavioral pathways. Economic factors shape attitude but not direct behavioral intention, suggesting that economic considerations influence through belief formation. Regulatory factors directly affect behavioral intention beyond attitude, suggesting that institutional legitimation provides independent behavioral motivation. Technical barriers suppress adoption directly, indicating that perceived complexity represents a behavioral deterrent independent from overall psychological orientation. These differential pathways align with recent research emphasizing the distinct mechanisms through which different factor categories influence technology adoption.

### 3.4 Environmental and Social Stimulus

The non-significant effects of environmental and social stimuli, contrary to H2 and H3, should be interpreted in relation to the socioeconomic and cultural context of Pulau Tidung rather than as evidence that these factors are inherently unimportant. First, immediate economic priorities may outweigh environmental considerations. Many households depend on fishing and tourism-related activities, both of which are subject to income variability and seasonal conditions. Under such circumstances, affordability, electricity expenditure, and the perceived economic consequences of solar PV adoption may receive greater attention than longer-term environmental benefits. Environmental awareness may therefore exist without becoming a sufficiently strong determinant of adoption-related attitudes when households face more immediate livelihood concerns.

Second, the weak social effect may reflect the characteristics of a relatively cohesive island community in which social influence operates differently from that observed in more mature solar PV markets. Pulau Tidung exhibits strong community interaction, yet households have limited exposure to visible solar PV adopters who could serve as credible peer examples. Consequently, conventional peer-effect mechanisms may remain weak because residents have relatively few opportunities to observe the practical benefits, reliability, and maintenance requirements of household solar PV systems. Informal lending and equipment-sharing practices within the community may further shift attention toward direct practical experience and perceived feasibility rather than generalized social approval. This interpretation is consistent with previous evidence showing that peer visibility and social influence become more influential when potential adopters are exposed to observable adoption examples (Alam et al., 2021; Graziano et al., 2019).

Third, limited awareness and market exposure may further suppress both environmental and social influences. The solar PV market in the Seribu Islands remains relatively nascent, with limited vendor presence, promotional activities, and opportunities for households to observe demonstration systems or interact with experienced adopters. These conditions restrict the information and social learning mechanisms through which environmental values and community influence might otherwise translate into stronger adoption attitudes. Accordingly, the non-significant environmental and social paths appear to reflect context-specific socioeconomic priorities, limited technology exposure, and the island community's particular social structure rather than the universal irrelevance of these determinants.

These non-significant relationships should nevertheless be interpreted cautiously given the sample size. The achieved sample provides greater sensitivity to moderate-to-large structural effects than to small effects; therefore, the lack of statistical significance should not be interpreted as definitive evidence that environmental or social influences are absent. Larger multi-site studies involving communities with different levels of solar PV exposure, market maturity, and socioeconomic characteristics are needed to determine whether these relationships become more pronounced under different contextual conditions.

### 3.5 Technical Barriers as Critical Adoption Constraint

The significant negative effect of technical barriers on adoption intention ($\beta = -0.306$, $p = 0.036$) represents a noteworthy finding. Respondents expressed concerns about installation complexity, maintenance cost

requirements, and system reliability despite high overall adoption willingness (89%). This paradox suggests that while desire to adopt solar energy exists, perceived capability to successfully implement and maintain systems creates substantial behavioral hesitation (Armitage & Conner, 2001). This pattern aligns with the TAM, which emphasizes that perceived ease of use represents a critical determinant of technology adoption independent from usefulness perceptions (Davis, 1989). Recent applications of TAM to renewable-energy adoption similarly confirm that perceived ease of use maintains strong predictive validity, even among populations demonstrating generally positive technology attitudes (Siuda & Gręboszk-Krawczyk, 2025). For distributed renewable energy adoption, technical barrier perceptions may particularly constrain behavior if users fear expensive repairs, worry about system failure during critical periods, or doubt local technician capability.

The present findings also provide a useful contrast with evidence from other developing-country and urban contexts. In Malaysia, Alam et al. (2021) reported that behavioral and social factors played important roles in household solar PV adoption. In contrast, social stimulus was not statistically significant in the present island-community setting. This difference may reflect variation in technology exposure, market maturity, and access to adopter networks. Similarly, studies conducted in more urbanized settings, such as Hong Kong, have emphasized policy support, market accessibility, and infrastructure-related considerations in shaping solar PV adoption (Mah et al., 2018). Compared with such urban environments, Pulau Tidung is characterized by greater geographic isolation, more limited vendor and technical-service availability, and stronger dependence on localized community networks. These contextual differences may help explain why technical barriers and regulatory support emerged as significant determinants in this study, while environmental, social, and market stimuli were comparatively weak. The findings therefore suggest that determinants of solar PV adoption should not be assumed to operate uniformly across urban, rural, and island communities, particularly in developing-country settings where infrastructure, service accessibility, and household economic priorities can differ substantially.

### 3.6 Practical Implications for Policy and Industry

The findings provide several actionable implications for policymakers. Given the significant positive effect of regulatory stimulus on adoption intention, policy support should move beyond general incentives toward clearly defined household-level mechanisms. These may include targeted upfront installation subsidies or capital-cost rebates, low-interest or subsidized credit for solar PV purchases, and net-metering arrangements that reduce the effective cost of adoption. For lower-income households, subsidies could be targeted according to household income or electricity expenditure to improve affordability while limiting unnecessary fiscal costs. Community-based financing and cooperative ownership models may also suit island settings, where individual households may struggle to bear the full initial investment (Jirakiattikul et al., 2021).

The significant negative effect of technical barriers indicates that financial incentives alone are unlikely to be sufficient. Local technical capacity should therefore be strengthened through a competency-based training and certification model for island-based technicians. Such a program could combine short theoretical modules on PV system operation and safety with supervised practical training in installation, routine maintenance, fault diagnosis, and basic component replacement. A train-the-trainer approach involving certified external specialists and selected local technicians could help establish sustainable technical capacity within the community. This should be complemented by local service points, scheduled preventive-maintenance programs, clear warranty arrangements, and access to remote technical assistance to reduce households' concerns regarding maintenance costs, system failure, and limited specialist availability.

For solar energy providers and development organizations, product and service packages should match island-community conditions. Pre-configured household systems, transparent lifecycle-cost information, extended warranties, and bundled installation-maintenance packages could reduce perceived complexity and uncertainty. Demonstration systems at schools, health facilities, or community centers could provide visible evidence of system reliability. At the same time, microfinance, cooperative purchasing, and energy service company (ESCO) arrangements could reduce upfront financial barriers. Together, these measures translate the empirical findings into targeted interventions that address the three principal mechanisms identified in this study: economic considerations, regulatory support, and perceived technical barriers (Abdul-Salam & Phimister, 2019).

### 3.7 Limitations and Future Research Directions

Several limitations should be considered when interpreting the findings. First, the relatively small sample size ($n$ = 100) limits the analysis's statistical sensitivity, particularly for detecting weak structural relationships. Although the retrospective PLS-SEM sensitivity assessment indicates that the achieved sample was adequate for the moderate-to-large significant relationships observed in this study, smaller effects may remain undetected and should therefore be interpreted cautiously. Future research should employ larger samples to improve statistical precision and the ability to identify weaker relationships.

Second, the use of non-probability purposive sampling limits external validity. Respondents were deliberately

selected because they represented household heads or primary energy decision-makers, which was appropriate for the specific research objective but does not provide a probability-based representation of the broader population. Consequently, interpret the findings as context-specific evidence from Pulau Tidung rather than population-wide estimates for Indonesian island communities. Replication using probability-based sampling, larger multi-site samples, and communities with different socioeconomic, infrastructural, and market conditions would strengthen the generalizability and external validity of the findings.

Third, the cross-sectional design captures household perceptions and adoption intentions at a single point in time and does not establish whether these intentions ultimately translate into actual solar PV adoption. In addition, reliance on self-reported questionnaire data may introduce social desirability, recall, and common method biases. Future longitudinal studies that combine survey responses with objective adoption records, administrative data, or observed household behavior would provide stronger evidence on the stability of the identified relationships and the transition from adoption intention to actual adoption.

Future research should therefore employ larger probability-based or multi-site samples, longitudinal designs, and complementary behavioral or administrative data to strengthen external validity and assess whether the identified determinants remain stable across different geographic and socioeconomic contexts. Comparative studies across multiple island communities would be particularly valuable for distinguishing context-specific effects from more general determinants of household solar PV adoption. In addition, the HTMT and Fornell–Larcker assessments indicated limited discriminant separation among several construct pairs, while both the inner-model and full-collinearity VIF diagnostics indicated substantial collinearity. Because all full-collinearity VIF values exceeded the conservative 3.3 benchmark, potential common method bias cannot be ruled out. These diagnostic concerns do not invalidate the estimated relationships, but they limit the extent to which individual structural coefficients can be interpreted as fully independent effects. Future studies should employ more differentiated measurement specifications, independent or multi-source data where feasible, and larger heterogeneous samples to reassess construct distinctiveness and the robustness of the identified relationships.

Future research should examine actual behavior, compare multiple island communities, investigate adoption barriers in depth, assess policy interventions, and explore gender dimensions. Track intention-behavior relationships longitudinally, investigating how post-adoption satisfaction affects subsequent renewable energy investment decisions and peer recommendations. Conduct multi-site studies identifying how geographic, socioeconomic, and market context differences influence adoption determinants, building on recent research emphasizing context-specific adoption patterns. Employ qualitative methods (interviews, focus groups) exploring detailed mechanisms through which technical, financial, and institutional barriers constrain behavior. Implement randomized controlled trials testing whether specific policy interventions (subsidies, training programs, vendor support) successfully translate adoption intentions into actual behavior. Investigate gender-specific adoption patterns, as this study's male-dominated sample may mask important gender differences in renewable energy decision-making (Mohandes et al., 2019).

## 4. Conclusions

This study examined the determinants of household solar PV adoption intention in Pulau Tidung, Indonesia, using an SOR-based framework informed by selected TPB and TAM concepts. The findings indicate that attitude toward solar PV exhibited the largest positive association with adoption intention ($\beta = 0.827$) among the examined relationships, suggesting that psychological evaluation may play an important role in household adoption decisions. Economic stimulus significantly strengthens attitude ($\beta = 0.411$), indicating that affordability, expected savings, and related financial considerations primarily influence adoption through attitude formation. Regulatory stimulus directly and positively influences adoption intention ($\beta = 0.349$), whereas perceived technical barriers exert a significant negative effect ($\beta = -0.306$). In contrast, environmental, social, and market stimuli do not demonstrate statistically significant effects within the hypothesized relationships. Collectively, these findings indicate that household solar PV adoption intention in an island setting is shaped by a combination of favorable attitudes, enabling institutional conditions, and perceived technical feasibility.

The study contributes to the renewable energy adoption literature in three respects. Theoretically, the study demonstrates how an SOR-based explanatory framework can be complemented by selected TPB and TAM concepts to interpret behavioral and technology-related mechanisms underlying household solar PV adoption intention. Empirically, it extends existing evidence by examining a developing-country island community, where geographic isolation, market accessibility, and technical capacity create adoption conditions that differ from those commonly studied in urban and developed-country settings. Practically, the findings suggest that policies aimed at accelerating household solar PV uptake should combine economically attractive schemes and clear regulatory support with accessible technical assistance and maintenance services. These insights provide a context-sensitive basis for designing renewable energy interventions in geographically constrained communities.

## Author Contributions

Conceptualization, H.M. and N.M.A.; methodology, H.M.; software, S.; validation, H.M., A.P.Y., and A.R.; formal analysis, H.M.; investigation, H.M.; resources, H.M.; data curation, H.M. and S.; writing—original draft preparation, H.M., N.M.A., S., A.P.Y., and A.R.; writing—review and editing, H.M. and A.P.Y.; visualization, S. and A.R.; supervision, H.M.; project administration, H.M. All authors have read and agreed to the published version of the manuscript.

## Informed Consent Statement

Participation was voluntary, and respondents were verbally informed about the purpose of the study, the nature of their participation, the confidentiality of their responses, and their right to decline participation. Verbal informed consent was obtained from all respondents before questionnaire administration.

## Ethical Approval

This study involved a non-interventional household survey. Formal ethical approval or an exemption determination was not obtained prior to data collection.

## Data Availability

The survey dataset is not publicly available and cannot be shared because it contains household-level information collected under conditions of respondent confidentiality. Access to the individual-level data is therefore restricted to protect participant privacy and to remain consistent with the confidentiality commitments made during data collection.

## Conflicts of Interest

The authors declare no conflicts of interest.

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